Operational Time Manipulation and Microarchitecture for Physical AGI

Baoliin (Zaitian) Wu 伍宝林

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1 Physical Proof of Existence

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2 Operational Time Manipulation and Microarchitecture for Physical AGI

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2.1 Abstract

Building on the novel definition of time, **time is the countable rhythm of Yin–Yang waxing and waning** in the companion paper DW58553925, this paper explores the engineering realization of **operational time manipulation** through observer-relative phase-locking control. The phrase *time manipulation* is used here strictly to mean the controlled adjustment of an internal oscillator' s phase offset, not the alteration of physical time coordinates or spacetime metrics. Based on this operational definition, we propose a principle-level control architecture for Physical Artificial General Intelligence (Physical AGI).

The geometric component begins with a real skew-symmetric matrix inspired by the Luoshu arrangement. Its matrix exponential is derived exactly as an element of $SO(3)$, yielding a mathematically verifiable rotation operator. We then distinguish this coordinate transformation from a projection in the strict mathematical sense and from a quantum measurement operation. A constrained mechanical model is formulated using Lagrange multipliers, with the equations of motion derived explicitly. The determinant-one condition is identified as a volume-preservation condition for a coordinate map, not as symplecticity; Liouville' s theorem is invoked only under its Hamiltonian-flow assumptions.

The proposed **Silicon Jade Cong** is a conceptual microarchitecture combining a geometric parameterization, phase-control logic, relative-coordinate exchange, and hardware-enforced safety interfaces. A Control Barrier Function (CBF) condition is stated together with the dynamical assumptions required for its use. A PyTorch implementation is provided as a Luoshu-inspired parameterized convolution prototype, not as a demonstrated $SO(3)$-equivariant neural operator. A Verilog interface skeleton identifies the required CORDIC dependency and safe default behavior; it is not presented as a functionally verified controller or completed synthesizable design. The paper therefore establishes a set of mathematical building blocks and an implementable research specification, while leaving system-level performance, safety, and physical efficacy for formal verification and experiment.

Keywords: Physical AGI; time manipulation; Yin–Yang phase-locking; Silicon Jade Cong; Luoshu-inspired geometry; geometric control; constrained dynamics; Control Barrier Functions; hardware safety

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2.2 1. Introduction: From Reading Time to Controlling Phase

The companion paper, *Time Is the Countable Rhythm of Yin–Yang Waxing and Waning* [1], develops an operational interpretation of time in terms of countable rhythms and observer-relative phase. The present paper explores one possible engineering consequence: whether an embodied intelligent system can use a controlled phase relationship as part of its internal timing, state estimation, and motion-control architecture.

The title of this paper deliberately employs the phrase *operational time manipulation* to highlight the engineering leap from the companion paper' s definitional architecture [1] to a physical controller. In the classical Chinese ontology reconstructed in [1], time is not an absolute container but the countable rhythm of Yin–Yang phase transitions. Therefore, *manipulating time* in this operational context does not mean changing an external physical time coordinate. Instead, it means **manipulating the phase relationship** between an internal oscillator (the system' s "Yin–Yang rhythm") and an observer-defined reference. By decoupling, shifting, and re-locking these phases, the proposed architecture effectively programs the system' s internal measure of time relative to its environment. This is the precise engineering analogue of the "Now" (*Ji* 几/幾) as a phase-locking reset point, transitioning from a philosophical definition to a controllable microarchitecture.

It is essential to distinguish this restricted operational sense from any claim of altering physical time, traveling through time, or changing the spacetime metric. Phase control is a standard engineering operation, whereas changing physical time would require a substantially different physical theory and experimental evidence.

The design is organized around four research questions:

1. Can a skew-symmetric generator inspired by the Luoshu arrangement define a mathematically verifiable rotation operator?
2. Can this operator be incorporated into a motion-control model with explicit state variables, constraints, and feedback?
3. Can the resulting functions be partitioned into a testable hardware architecture?
4. Under what assumptions might the geometric parameterization improve localization, safety, or control, and which performance claims require experimental validation?

The paper is a conceptual and mathematical engineering proposal. It does not report a fabricated chip, a completed ASIC, a validated Physical AGI system, or experimental evidence of improved localization or safety. The terms *Physical AGI* and *Silicon Jade Cong* name the proposed research direction and architecture.

The remainder of the paper is organized as follows. Section 2 derives the Luoshu-inspired rotation generator. Section 3 defines the scope and limits of the geometric state transformation. Section 4 develops phase dynamics, constrained equations of motion, and a CBF safety condition. Section 5 describes the proposed microarchitecture and relative-coordinate alignment. Section 6 discusses hardware-constrained safety and ethical specifications. Section 7 summarizes the contribution and the validation work still required. Appendices A and B provide preliminary software and RTL artifacts.

2.3 2. Theoretical Foundation: A Luoshu-Inspired Rotation Generator

2.3.1 2.1 Definition of the generator

Consider the real matrix

$$A = \begin{pmatrix} 0 & 3 & -3 \\ -3 & 0 & 3 \\ 3 & -3 & 0 \end{pmatrix} \quad (1)$$

It is skew-symmetric: $A^T = -A$. Direct multiplication gives

$$A^2 = \begin{pmatrix} -18 & 9 & 9 \\ 9 & -18 & 9 \\ 9 & 9 & -18 \end{pmatrix}, \qquad A^3 = -27A. \quad (2)$$

The matrix is therefore a scaled cross-product generator. Under the convention

$$[\mathbf{n}]_\times = \begin{pmatrix} 0 & -n_3 & n_2 \\ n_3 & 0 & -n_1 \\ -n_2 & n_1 & 0 \end{pmatrix},$$

we have

$$A = 3\sqrt{3}\,[\hat{\mathbf{n}}]_\times, \qquad \hat{\mathbf{n}} = -\frac{1}{\sqrt{3}}(1,1,1)^T.$$

Thus $\hat{\mathbf{n}}$ is a unit rotation axis. The connection to the Luoshu is a proposed algebraic construction: the matrix in Eq. (1) is the antisymmetric generator used in this model. It should not be interpreted as a derivation of this matrix from a unique historical or canonical Luoshu transformation unless the precise centering, ordering, and antisymmetrization map is separately specified.

2.3.2 2.2 Exact exponential map

Let θ be a dimensionless rotation parameter and define $\phi = 3\sqrt{3}\theta$. Because $A^3 = -27A$, higher powers of A reduce to linear combinations of A and A^2. Expanding the matrix exponential and collecting even and odd powers gives

$$R(\theta) = \exp(\theta A) = I + \frac{\sin\phi}{3\sqrt{3}}A + \frac{1-\cos\phi}{27}A^2. \quad (3)$$

Since $A^T = -A$,

$$R(\theta)^T = \exp(\theta A^T) = \exp(-\theta A) = R(\theta)^{-1}.$$

Consequently,

$$R(\theta)^T R(\theta) = I, \qquad \det R(\theta) = 1, \qquad R(\theta) \in SO(3). \quad (4)$$

The operator preserves Euclidean vector norms, angles, and orientation. These properties follow from the skew-symmetry of A; they do not depend on an empirical claim about the performance of a physical device.

The rotation axis may be used as a mathematical orientation reference in a controller. The comparison with a plumb line in discussions of the *Zhoubi Suanjing* is an interpretive analogy, not an assertion that the historical instrument and the present matrix are mathematically or physically identical.

2.4 3. Geometric State Representation: Scope and Mathematical Boundaries

2.4.1 3.1 Coordinate transformation, not automatically a projection

Let $\mathbf{x} \in \mathbb{R}^3$ be a classical state vector in a specified coordinate chart. Let $C_{\text{Cong}} \in \mathbb{R}^{3\times 3}$ be a calibration or coordinate-mapping matrix. The proposed geometric transformation is

$$\mathbf{x}_{\text{obs}} = C_{\text{Cong}} R(\theta)\mathbf{x}. \quad (5)$$

Here $R(\theta)$ supplies a rotation and C_{Cong} maps the rotated coordinates into the chosen observer or device frame. If C_{Cong} is invertible, Eq. (5) is an invertible coordinate transformation. If it is non-invertible or includes quantization, the resulting operation may lose information; its properties must then be analyzed separately.

The name *Coordinate-Forced Projection* may be used as a project-specific label for the proposed observer-conditioned mapping, but Eq. (5) is not, by itself, a projection operator in the strict linear-algebraic sense. A linear projection P must satisfy $P^2 = P$. A generic rotation R does not satisfy this condition, except in special cases. A continuous-to-discrete mapping would additionally require a defined quantizer, codebook, lattice, or measurement map. No such discrete map is assumed in Eq. (5).

The Luoshu arrangement therefore motivates a geometric parameterization in this paper; it does not yet provide a mathematically complete construction of a discrete Luoshu state lattice.

2.4.2 3.2 Boundary between classical coordinates and quantum states

Equation (5) acts on a three-component classical vector. It cannot be applied directly to an arbitrary quantum state unless a Hilbert-space representation and a corresponding operator are specified. In particular, a 3×3 real rotation matrix, a componentwise product, and an unspecified gauge factor do not define a general quantum measurement rule.

A quantum extension would need, at minimum:

1. a Hilbert space $\mathcal{H}$ and a specified representation of the rotation group on $\mathcal{H}$;
2. a unitary operator $U(\theta)$ acting on that space;
3. a defined state-update or measurement instrument, including its measurement operators;
4. a probability rule for measurement outcomes; and
5. a physical account of how the device couples to the measured system.

For a unitary representation, the transformation would take the form

$$|\psi'\rangle = U(\theta)|\psi\rangle, \qquad U(\theta)^\dagger U(\theta) = I. \quad (6)$$

Equation (6) is a generic formal requirement, not a derived MES or Luoshu quantum operator. The present manuscript does not claim to solve the quantum measurement problem. The phrase *Now* or *Ji* 丌 is retained as a philosophical description of a transition threshold in an observer' s operational record; a physical model of that threshold would require a specified dynamical and measurement process.

2.5 4. The Phase-Locking Engine and Constrained Dynamics

2.5.1 4.1 Scalar phase dynamics

The phase-locking engine is proposed as a control loop with three functional stages:

1. **Decoupling:** temporarily reduce or suspend coupling to the reference oscillator;
2. **Phase conversion:** apply a controlled phase adjustment;
3. **Re-locking:** restore coupling and reduce the phase error relative to the reference.

A minimal scalar phase model is

$$\dot{\phi}(t) = \omega_0 + \alpha q(t), \quad (7)$$

where ϕ is an unwrapped phase in radians, ω_0 is the free-running angular frequency in s^{-1}, $q(t)$ is a dimensionless control or observable signal, and α has units of s^{-1}. Its integral form is

$$\phi(t) = \phi(0) + \omega_0 t + \alpha \int_0^t q(\tau)\, d\tau. \quad (8)$$

The reported phase on a circle is obtained by reducing the unwrapped phase modulo 2π. This distinction matters when accumulating phase over multiple cycles.

For a reference phase $\phi_{\mathrm{ref}}(t)$, define the wrapped phase error

$$e_\phi(t) = \mathrm{wrap}_{(-\pi,\pi]}\left(\phi(t) - \phi_{\mathrm{ref}}(t)\right).$$

A simple feedback law could be specified as

$$q(t) = \mathrm{sat}_{[q_{\min}, q_{\max}]}\left(-k_\phi e_\phi(t)\right), \quad k_\phi > 0, \quad (9)$$

where sat denotes actuator or command saturation. This is a candidate proportional phase-correction law, not a complete phase-locked-loop design. Stability and acquisition performance depend on the oscillator model, reference dynamics, sampling, delays, noise, saturation, and the selected gain. A practical design must also specify the decoupling and re-locking control schedule.

If the phase conversion at the k-th cycle is $\Delta\Phi_{\text{conv}}^{(k)}$, the accumulated commanded phase offset is

$$\Delta\Phi_{\text{total}} = \sum_{k=1}^{N} \Delta\Phi_{\text{conv}}^{(k)}. \quad (10)$$

This quantity is a phase offset, not an elapsed-time displacement or a change in the physical time coordinate.

2.5.2 4.2 Lagrangian model and constraint definitions

Let $\mathbf{x} \in \mathbb{R}^n$ be generalized coordinates, $M \in \mathbb{R}^{n\times n}$ a constant symmetric positive-definite mass matrix, and $V(\mathbf{x}, t)$ a differentiable potential. Define

$$L(\mathbf{x}, \dot{\mathbf{x}}, t) = \frac{1}{2}\dot{\mathbf{x}}^T M \dot{\mathbf{x}} - V(\mathbf{x}, t). \quad (11)$$

The following constraints are proposed as examples of geometric and kinematic requirements:

Kinematic or spin-reference constraint

$$\mathbf{g}_1(\mathbf{x}, \dot{\mathbf{x}}) = J_{\text{Cong}}\dot{\mathbf{x}} - \mathbf{v}_{\text{ref}} = \mathbf{0},$$

where $J_{\text{Cong}} \in \mathbb{R}^{m\times n}$ is a specified, constant Jacobian-like mapping and $\mathbf{v}_{\text{ref}} \in \mathbb{R}^m$ is a reference velocity.

Coordinate-map volume constraint

$$g_2(\mathbf{x}) = \det\big(\nabla_{\mathbf{x}} C_{\text{Cong}}(\mathbf{x})\big) - 1 = 0.$$

This scalar constraint is meaningful only when $C_{\text{Cong}}(\mathbf{x})$ is differentiable and its Jacobian is square. If C_{Cong} is a constant matrix, its determinant condition is a parameter constraint rather than a state-dependent constraint, and $\nabla_{\mathbf{x}} g_2 = 0$.

Height or vertical-reference constraint

$$g_3(\mathbf{x}) = \mathbf{x} \cdot \hat{\mathbf{e}}_z - h_{\text{ref}} = 0,$$

where $\hat{\mathbf{e}}_z$ is a defined unit vertical direction and h_{ref} is the target coordinate along that direction.

For compactness, form the augmented Lagrangian

$$L_c = L + \lambda_1^T \mathbf{g}_1 + \lambda_2 g_2 + \lambda_3 g_3,$$

where $\lambda_1 \in \mathbb{R}^m$ and $\lambda_2, \lambda_3 \in \mathbb{R}$. The scalar nature of g_3 means that its multiplier is a scalar.

2.5.3 4.3 Euler–Lagrange equation with multipliers

Assume M and J_{Cong} are constant and $\mathbf{v}_{\text{ref}}$ is independent of $\mathbf{x}$. Then

$$\frac{\partial L_c}{\partial \dot{\mathbf{x}}} = M\dot{\mathbf{x}} + J_{\text{Cong}}^T \lambda_1,$$

and

$$\frac{d}{dt}\left(\frac{\partial L_c}{\partial \dot{\mathbf{x}}}\right) = M\ddot{\mathbf{x}} + J_{\text{Cong}}^T \dot{\lambda}_1.$$

The coordinate derivative is

$$\frac{\partial L_c}{\partial \mathbf{x}} = -\nabla_{\mathbf{x}} V + \lambda_2 \nabla_{\mathbf{x}} g_2 + \lambda_3 \hat{\mathbf{e}}_z.$$

The Euler–Lagrange equations therefore give

$$M\ddot{\mathbf{x}} + \nabla_{\mathbf{x}} V = -J_{\text{Cong}}^T \dot{\lambda}_1 + \lambda_2 \nabla_{\mathbf{x}} g_2 + \lambda_3 \hat{\mathbf{e}}_z. \quad (12)$$

Equation (12) is the constrained equation of motion for the stated model and sign convention. It is not, by itself, a closed numerical controller: the constraint equations, multiplier evolution or solution procedure, initial conditions, and any actuator dynamics must also be supplied. If J_{Cong} depends on $\mathbf{x}$, additional coordinate-derivative terms appear. If the constraints are imposed through a numerical optimization or differential-algebraic-equation solver, that implementation must be specified.

The term *Master Control Equation* may be used as a project label for Eq. (12), but the equation is a standard multiplier-based constrained Euler–Lagrange form under the assumptions stated above. It does not establish that every trajectory is a geodesic of an unspecified Riemannian manifold. A geodesic interpretation requires a defined metric and a compatible connection.

2.5.4 4.4 Volume preservation and symplectic structure

The condition $\det J_C = 1$, where $J_C = \nabla_{\mathbf{x}} C_{\text{Cong}}$, states that the coordinate map is locally volume-preserving, assuming a differentiable square map and the appropriate orientation. It is not, by itself, a symplectic condition.

For a transformation F on an even-dimensional phase space with canonical symplectic matrix Ω, symplecticity requires

$$(DF)^T \Omega (DF) = \Omega. \quad (13)$$

A symplectic map preserves the symplectic form and consequently phase-space volume, but the converse is not generally true. Therefore, a determinant-one condition on a coordinate map must not be described as symplecticity unless Eq. (13) is separately established on the relevant phase space.

Liouville's theorem states that Hamiltonian flow preserves phase-space volume. Under Hamiltonian dynamics,

$$\frac{d}{dt}\, d\Gamma = 0, \quad (14)$$

where $d\Gamma$ is the infinitesimal phase-space volume element. This result does not, by itself, guarantee numerical stability, state-estimation accuracy, model fidelity, or zero representational drift. It also does not automatically follow from the constrained model in Eq. (12); the relevant Hamiltonian structure and assumptions must be demonstrated.

2.5.5 4.5 Dynamic obstacle avoidance using a Control Barrier Function

Let the controlled dynamics be $\dot{\mathbf{x}} = f(\mathbf{x}, t) + B(\mathbf{x}, t)\mathbf{u}$, where $\mathbf{u}$ is the control input. Define a safe set $\mathcal{S}(t) = \{\mathbf{x} : h(\mathbf{x}, t) \geq 0\}$, where h is continuously differentiable. A common time-varying CBF condition is

$$\frac{\partial h}{\partial t} + \nabla_{\mathbf{x}} h^T \left(f + B\mathbf{u}\right) + \gamma h(\mathbf{x}, t) \geq 0, \qquad \gamma > 0. \quad (15)$$

A controller can seek an admissible $\mathbf{u}$ satisfying this inequality while minimizing deviation from a nominal command. The forward-invariance conclusion is conditional: the initial state must lie in the safe set; the model and obstacle estimates must be sufficiently accurate; an admissible input must exist; and sampling, computation, communication, and actuator delays must be accounted for. Input bounds, actuator saturation, sensor faults, and emergency-stop behavior must also be included in a real implementation.

Thus, the CBF provides a formal safety-control condition under stated assumptions. It is not a guarantee of collision avoidance in every physical environment, and it does not follow from the volume-preservation property in Eq. (14).

2.6 5. The Silicon Jade Cong Microarchitecture

2.6.1 5.1 Functional architecture

The **Silicon Jade Cong** is proposed as a modular controller architecture rather than a completed chip. Its functional blocks are:

1. **Reference and sensing interface:** receives inertial, visual, ranging, or other sensor data and maintains calibration metadata.
2. **Geometric transform unit:** evaluates the Luoshu-inspired rotation $R(\theta)$ and applies the selected coordinate map.
3. **Phase-control unit:** estimates phase error and generates bounded phase-correction commands.
4. **Constrained-dynamics and planning unit:** evaluates nominal trajectories and constraint residuals.
5. **Safety supervisor:** checks CBF inequalities, command bounds, watchdog status, and fault conditions.

6. **Actuator interface:** issues commands only when the safety supervisor and hardware interlocks permit them.
7. **Multi-agent communication interface:** exchanges relative pose and timing information with neighboring agents.

This partition is a proposed architecture. The paper does not claim that the complete design has been synthesized, placed, routed, fabricated, or benchmarked.

2.6.2 5.2 Gravity reference and observability

A gravity sensor can provide a local vertical reference under quasi-static or suitably compensated conditions:

$$\hat{\mathbf{e}}_z \approx \frac{\mathbf{g}_{\text{meas}}}{\|\mathbf{g}_{\text{meas}}\|}. \quad (16)$$

In practice, an accelerometer measures specific force, which includes contributions from platform acceleration, vibration, bias, and noise. The gravity direction therefore cannot always be inferred from a single instantaneous accelerometer sample. Filtering, sensor fusion, calibration, and motion-state estimation may be required.

A gravity reference can support roll and pitch estimation under suitable sensing and motion conditions. It does not determine yaw about the gravity vector. Heading generally requires an additional reference, such as a magnetometer, visual landmarks, GNSS heading, or another external orientation source. Under constrained planar motion, an $SE(2)$ model may be appropriate, provided the assumptions of planar motion and known vertical are satisfied.

2.6.3 5.3 Juedi Tiantong: relative-coordinate alignment

For two agents A and B, a candidate relative-coordinate transformation is

$$\mathbf{x}_B = C^B_{\text{Cong}} \left[R_{\text{Luoshu}}(\Delta\phi)(C^A_{\text{Cong}})^{-1}\mathbf{x}_A + \mathbf{T}_{AB} \right], \quad (17)$$

where $\mathbf{T}_{AB}$ is a relative translation estimated from available observations and $\Delta\phi$ is a relative orientation or phase parameter. This expression is meaningful only when the coordinate maps are invertible and the frames, units, and state representations are consistently defined.

Equation (17) is a relative-coordinate transformation. It is not, by itself, a distributed-consensus algorithm. A consensus claim would require a communication graph, agent state variables, neighbor-exchange rules, an update law, and convergence conditions, including treatment of communication delay, packet loss, and clock mismatch. The present proposal therefore uses *Juedi Tiantong* to describe decentralized relative-coordinate alignment, not a proven global-consensus protocol.

The phrase *Juedi Tiantong* (severing communication between Earth and Heaven) is used here as a conceptual metaphor for establishing controlled observer-relative reference frames. Its interpretation is connected to the companion work on countable Yin–Yang rhythms [1], coordinate representation [14], and the Gui-Ju geometric paradigm [2, 15]. These connections are interpretive and programmatic; they do not establish historical equivalence between ancient institutions and modern control theory.

Relative alignment can reduce dependence on a single global positioning reference in some configurations, but it does not eliminate accumulated drift. Its accuracy depends on sensor quality, network topology, observability, calibration, and the estimation algorithm.

2.6.4 5.4 Instruction set proposal

The proposed command set is:

Opcode	Instruction	Intended function
001	PING_INIT	Initialize sensing and reference checks
010	LUOSHU_ROT	Request a Rodrigues rotation parameter update
011	HETU_TRANS	Request a coordinate translation
100	MUL_PUMP_ACT	Request a constrained actuation update

These names define a proposed instruction-level interface, not a finalized ISA specification. A complete implementation must define command encoding, operand widths, signedness, fixed-point formats, latency, valid/ready handshakes, exception behavior, reset semantics, and safety interlocks.

2.6.5 5.5 Hardware verification plan

Before the design can be described as a verified controller, the following stages are required:

- **Mathematical reference tests:** compare fixed-point rotation outputs against a high-precision software implementation.
- **Property tests:** verify orthogonality and determinant bounds within a declared numerical tolerance.
- **Protocol tests:** verify command sequencing, valid/ready timing, reset, timeout, and invalid-opcode behavior.
- **Arithmetic tests:** verify signed multiplication, rounding, saturation, overflow detection, and unit scaling.
- **Safety tests:** verify that invalid sensor states, CBF infeasibility, watchdog timeout, and actuator faults force a defined safe output.
- **Formal properties:** specify and prove relevant invariants for the implemented RTL.
- **Physical validation:** measure latency, power, area, sensor performance, and closed-loop behavior on the intended platform.

2.7 6. Hardware-Constrained Safety and Ethical Specifications

2.7.1 6.1 Three-layer safety architecture

The proposed safety architecture has three layers:

1. **Specification layer:** defines explicit safe sets, forbidden states, operating envelopes, and permitted actuator ranges.
2. **Control layer:** computes nominal commands and applies CBF-based constraints or other verified control filters.
3. **Hardware layer:** independently gates actuator outputs, monitors watchdogs and fault signals, and enters a defined safe state when required.

This architecture can enforce specified constraints if the sensors, logic, actuators, and failure assumptions are correctly implemented. It cannot, by hardware alone, resolve every open-ended ethical judgment or guarantee that the specification itself is complete and appropriate.

2.7.2 6.2 Barrier potentials and multiplier distinction

A steep potential may be used to discourage approach to a designated boundary. For example, for a scalar distance $d(\mathbf{x}) > 0$ to a boundary, one may define an idealized barrier potential

$$V_{\text{barrier}}(\mathbf{x}) = \frac{k}{d(\mathbf{x})^p}, \qquad k > 0, \quad p > 0. \quad (18)$$

This expression diverges as $d \to 0^+$, but a physical controller has finite numerical precision, finite actuator authority, and finite response time. The divergence is therefore a mathematical idealization, not a physical guarantee that a boundary can never be crossed.

The multiplier λ_2 in Section 4 is associated with the coordinate-map volume constraint g_2. It is not defined as an ethical-risk signal and must not be interpreted as automatically increasing near an ethical boundary. If a constrained optimization-based safety filter is used, its safety-related multipliers should be defined separately. For example, a nonnegative optimization multiplier μ_{safe} may be associated with a CBF inequality in a quadratic program. Its value and interpretation depend on the specific optimization problem.

2.7.3 6.3 Limits of hardware alignment

A hardware interlock can enforce explicit conditions such as current limits, velocity limits, restricted workspace, or emergency-stop commands. It cannot determine by itself whether all relevant human values have been represented correctly. Ethical alignment therefore requires specification review, operational governance, monitoring, auditability, and procedures for updating the safety model. The proposed Silicon Jade Cong should be understood as a possible technical layer within a broader safety system, not as a complete solution to AI alignment.

2.8 7. Conclusion

This paper has proposed a principle-level microarchitecture for Physical AGI centered on observer-relative phase control and a Luoshu-inspired geometric parameterization. The skew-symmetric generator in Eq. (1) yields the exact Rodrigues rotation in Eq. (3), with the expected $SO(3)$ properties. This is the manuscript' s principal closed mathematical result.

The paper has also clarified the boundaries of the proposed architecture. The classical coordinate transformation is not automatically a discrete projection or a quantum measurement operation. The determinant-one condition expresses local volume preservation, not symplecticity. Liouville' s theorem applies under Hamiltonian-flow assumptions and does not imply zero numerical or representational drift. The constrained Euler–Lagrange equation is valid for the stated constant-matrix assumptions, but a complete controller still requires constraint solving, feedback design, and actuator models. The CBF condition provides conditional forward-invariance requirements rather than an unconditional physical safety guarantee.

The Silicon Jade Cong is therefore presented as a research architecture and implementation roadmap. The PyTorch and Verilog appendices are preliminary prototypes; neither demonstrates a complete, validated Physical AGI system. Future work should define the discrete state map, develop a full phase-locking controller, specify the distributed estimation algorithm, implement and verify the fixed-point arithmetic and CORDIC interface, and evaluate the system on hardware under reproducible test conditions.

The proposed connection between Yin–Yang phase concepts, Luoshu-inspired geometry, and embodied computation remains a research hypothesis. Its value will depend on whether it yields measurable improvements over appropriate baseline systems in phase stability, state-estimation error, control performance, safety, latency, power consumption, and robustness.

2.9 Appendix A: PyTorch Luoshu-Inspired Parameterized Convolution Prototype

The following implementation constructs a 3×3 convolution kernel from the Rodrigues matrix and a Gaussian envelope. It is a parameterized convolution prototype inspired by the generator in Section 2. The mapping of the nine matrix entries to spatial kernel locations is a design choice; it does not, by itself, establish $SO(3)$-equivariance, physical rotation of an input field, or a validated geometric neural operator.

```python
import math

import torch
import torch.nn as nn
import torch.nn.functional as F


class LuoshuRodriguesConv2d(nn.Module):
    """
    Luoshu-inspired parameterized 3x3 convolution prototype.

    The Rodrigues matrix entries parameterize a spatial kernel.
    This construction is not claimed to be SO(3)-equivariant.
    """

    def __init__(
        self,
        in_channels: int,
        out_channels: int,
        kernel_size: int = 3,
        sigma: float = 1.0,
    ):
        super().__init__()

        if kernel_size != 3:
            raise ValueError("This prototype uses a fixed 3x3 kernel.")
```

```
        if sigma <= 0:
            raise ValueError("sigma must be strictly positive.")
        if in_channels <= 0 or out_channels <= 0:
            raise ValueError("Channel counts must be positive.")

        self.in_channels = in_channels
        self.out_channels = out_channels
        self.sigma = float(sigma)

        A = torch.tensor(
            [
                [0.0, 3.0, -3.0],
                [-3.0, 0.0, 3.0],
                [3.0, -3.0, 0.0],
            ],
            dtype=torch.float32,
        )
        self.register_buffer("A", A)
        self.register_buffer("A_sq", A @ A)

        # One learnable phase parameter for each input/output channel pair.
        self.theta = nn.Parameter(
            torch.randn(out_channels, in_channels) * 0.1
        )

        grid_y, grid_x = torch.meshgrid(
            torch.linspace(-1.0, 1.0, 3),
            torch.linspace(-1.0, 1.0, 3),
            indexing="ij",
        )
        envelope = torch.exp(
            -(grid_x.square() + grid_y.square()) / (2.0 * sigma**2)
        )
        envelope = envelope / envelope.sum()
        self.register_buffer("gaussian_envelope", envelope)

    def forward(self, x: torch.Tensor) -> torch.Tensor:
        if x.ndim != 4:
            raise ValueError("Input must have shape (N, C, H, W).")
        if x.shape[1] != self.in_channels:
            raise ValueError(
                f"Expected {self.in_channels} input channels, "
                f"received {x.shape[1]}."
            )

        # Match buffers to the input dtype/device for mixed-device safety.
        A = self.A.to(device=x.device, dtype=x.dtype)
        A_sq = self.A_sq.to(device=x.device, dtype=x.dtype)
```

```
        envelope = self.gaussian_envelope.to(
            device=x.device, dtype=x.dtype
        )
        theta = self.theta.to(device=x.device, dtype=x.dtype)

        phi = 3.0 * math.sqrt(3.0) * theta
        sin_phi = torch.sin(phi)
        cos_phi = torch.cos(phi)

        identity = torch.eye(
            3, device=x.device, dtype=x.dtype
        ).view(1, 1, 3, 3)

        term_a = (
            sin_phi / (3.0 * math.sqrt(3.0))
        ).view(self.out_channels, self.in_channels, 1, 1) * (
            A.view(1, 1, 3, 3)
        )

        term_a_sq = (
            (1.0 - cos_phi) / 27.0
        ).view(self.out_channels, self.in_channels, 1, 1) * (
            A_sq.view(1, 1, 3, 3)
        )

        rotation_parameter = identity + term_a + term_a_sq
        filters = rotation_parameter * envelope.view(1, 1, 3, 3)

        return F.conv2d(x, filters, padding=1)


def rodrigues_matrix(theta: torch.Tensor) -> torch.Tensor:
    """Return the 3x3 Rodrigues matrix for a scalar theta."""
    A = torch.tensor(
        [
            [0.0, 3.0, -3.0],
            [-3.0, 0.0, 3.0],
            [3.0, -3.0, 0.0],
        ],
        dtype=theta.dtype,
        device=theta.device,
    )
    phi = 3.0 * math.sqrt(3.0) * theta
    I = torch.eye(3, dtype=theta.dtype, device=theta.device)
    return (
        I
        + (torch.sin(phi) / (3.0 * math.sqrt(3.0))) * A
        + ((1.0 - torch.cos(phi)) / 27.0) * (A @ A)
```

```
    )


def test_rodrigues_matrix():
    theta = torch.tensor(0.37, dtype=torch.float64)
    R = rodrigues_matrix(theta)
    I = torch.eye(3, dtype=R.dtype, device=R.device)
    assert torch.allclose(R.T @ R, I, atol=1e-10, rtol=1e-10)
    assert torch.allclose(
        torch.linalg.det(R),
        torch.ones((), dtype=R.dtype, device=R.device),
        atol=1e-10,
        rtol=1e-10,
    )


if __name__ == "__main__":
    test_rodrigues_matrix()

    model = LuoshuRodriguesConv2d(
        in_channels=3,
        out_channels=8,
        sigma=1.0,
    )
    sample = torch.randn(2, 3, 32, 32)
    output = model(sample)
    print("Output shape:", tuple(output.shape))
```

The prototype should be tested for forward and backward execution, dtype/device compatibility, gradient behavior, numerical stability, and agreement with an independent reference implementation. Additional tests are needed before using it in a safety-critical control path.

2.10 Appendix B: Preliminary Verilog RTL Interface Skeleton

The following RTL illustrates command handling, safe reset values, a gravity-reference check, and an explicit external CORDIC interface. The CORDIC module is an external dependency and must be supplied and verified separately. The actuation output is deliberately held at zero because this appendix does not yet implement a validated constraint-force arithmetic path.

The code is an interface-level RTL skeleton. It is not a completed controller, and synthesis success or functional correctness has not been established.

```
// ============================================================================
// Module: silicon_jade_cong_core
// Purpose: Preliminary interface skeleton for the Silicon Jade Cong.
// Fixed-point convention: signed Q16.16 unless otherwise specified.
// External dependency: cordic_sincos_q16.
// ============================================================================
```

```
module silicon_jade_cong_core (
    input  wire                clk,
    input  wire                rst_n,

    input  wire                cmd_valid,
    input  wire [2:0]          cmd_opcode,
    input  wire signed [31:0]  cmd_param_theta,  // Q16.16 phase parameter

    input  wire signed [31:0]  gravity_vector_z, // calibrated Q16.16 input

    output reg                 core_ready,
    output reg                 z_axis_locked,
    output reg  signed [31:0]  r_sin_phi,        // Q16.16
    output reg  signed [31:0]  r_cos_phi,        // Q16.16
    output reg  signed [31:0]  actuation_pwm_out // Q16.16; safe default = 0
);

    // Correctly rounded Q16.16 constants.
    // 3*sqrt(3) = 5.1961524227...
    // 1/(3*sqrt(3)) = 0.1924500897...
    // 1/27 = 0.0370370370...
    localparam signed [31:0] CONST_THREE_SQRT_THREE = 32'h00053237;
    localparam signed [31:0] INV_THREE_SQRT_THREE   = 32'h00003144;
    localparam signed [31:0] INV_TWENTY_SEVEN       = 32'h0000097B;

    // Example positive-Z threshold of approximately 0.9 in Q16.16.
    // This assumes a calibrated, signed sensor value and a known axis convention.
    localparam signed [31:0] GRAVITY_Z_MIN = 32'sh0000E666;

    wire signed [31:0] cordic_sin;
    wire signed [31:0] cordic_cos;
    wire               cordic_valid;

    reg                cordic_start;
    reg signed [31:0]  cordic_phase;
    wire signed [63:0] theta_scaled_full;
    wire signed [31:0] phi_q16;

    // Convert theta to phi = (3*sqrt(3))*theta in Q16.16.
    assign theta_scaled_full = cmd_param_theta * CONST_THREE_SQRT_THREE;
    assign phi_q16 = theta_scaled_full >>> 16;

    // External module contract:
    // - phase_in receives phi = (3*sqrt(3))*theta in Q16.16.
    // - sin_out and cos_out are Q16.16.
    // - valid_out pulses when outputs are valid.
    // The implementation must define phase range, latency, rounding,
```

```
// overflow behavior, and approximation error.
cordic_sincos_q16 u_cordic (
    .clk       (clk),
    .rst_n     (rst_n),
    .start     (cordic_start),
    .phase_in  (cordic_phase),
    .sin_out   (cordic_sin),
    .cos_out   (cordic_cos),
    .valid_out (cordic_valid)
);

localparam [2:0] OP_PING_INIT = 3'b001;
localparam [2:0] OP_LUOSHU_ROT = 3'b010;
localparam [2:0] OP_HETU_TRANS = 3'b011;
localparam [2:0] OP_MUL_PUMP_ACT = 3'b100;

always @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        core_ready       <= 1'b1;
        z_axis_locked    <= 1'b0;
        r_sin_phi        <= 32'sd0;
        r_cos_phi        <= 32'sh00010000; // +1.0 in Q16.16
        actuation_pwm_out <= 32'sd0;
        cordic_start     <= 1'b0;
        cordic_phase     <= 32'sd0;
    end else begin
        // Default pulse behavior.
        cordic_start <= 1'b0;

        // Accept a new command only while the interface is ready.
        if (cmd_valid && core_ready) begin
            case (cmd_opcode)

                OP_PING_INIT: begin
                    core_ready <= 1'b0;

                    // This simple threshold is only an example.
                    // A real design must validate magnitude, sign,
                    // calibration, motion state, and sensor health.
                    if (gravity_vector_z >= GRAVITY_Z_MIN) begin
                        z_axis_locked <= 1'b1;
                        core_ready    <= 1'b1;
                    end else begin
                        z_axis_locked <= 1'b0;
                        core_ready    <= 1'b1;
                    end
                end
```

```
                    OP_LUOSHU_ROT: begin
                        if (z_axis_locked) begin
                            cordic_phase <= phi_q16;
                            cordic_start <= 1'b1;
                            core_ready   <= 1'b0;
                        end
                    end

                    OP_HETU_TRANS: begin
                        // Coordinate translation datapath is not implemented.
                        // Keep the actuator output in its safe state.
                        actuation_pwm_out <= 32'sd0;
                    end

                    OP_MUL_PUMP_ACT: begin
                        // Constraint-force arithmetic is not implemented.
                        // Do not forward an unverified value to an actuator.
                        actuation_pwm_out <= 32'sd0;
                    end

                    default: begin
                        // Invalid opcode: preserve safe output and readiness.
                        actuation_pwm_out <= 32'sd0;
                        core_ready <= 1'b1;
                    end
                endcase
            end

            // Capture CORDIC results only when the external unit asserts valid.
            if (cordic_valid) begin
                r_sin_phi <= cordic_sin;
                r_cos_phi <= cordic_cos;
                core_ready <= 1'b1;
            end
        end
    end

endmodule
```

2.10.1 B.1 RTL implementation requirements

Before integration, the external CORDIC block and the surrounding controller must specify and verify:

- the phase input range and wrap convention;
- the exact Q-format of every port and intermediate result;
- CORDIC iteration count, gain compensation, rounding, and maximum error;
- valid/ready behavior and worst-case latency;

- reset and timeout behavior if `cordic_valid` never arrives;
- sensor calibration, sign convention, and fault detection;
- arithmetic overflow and saturation behavior;
- a complete, independently verified actuator safety gate.

The constants above are Q16.16 representations of the stated mathematical values. The skeleton includes the Q16.16 conversion from the command parameter theta to the Rodrigues angle phi. A production datapath must still verify signed multiplication, rounding, saturation, overflow handling, and the CORDIC input range.

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2.12 Funding

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2.13 Availability of data and material

Code, mathematical scripts, and preprints are available at GitHub: https://github.com/Zaitian001/-Self-Preprint-V1.1-

2.14 Competing interests

Not applicable.

2.15 Author Contributions

Baoliin (Zaitian) Wu 伍宝林 is the sole author of this manuscript and is responsible for all aspects of the research, including conceptualization, methodology, theoretical formulation, and manuscript writing.

2.16 References

[1] Baoliin (Zaitian) Wu 伍宝林. (2026). Time Is the Countable Rhythm of Yin–Yang Waxing and Waning. Self-Preprint Archive. https://Zaitian001.github.io/-Self-Preprint-V1.1-/preprints/DW58553925.html

[2] Baoliin (Zaitian) Wu 伍宝林. (2026). The Gui-Ju Dao: The Artificial General Intelligence Algorithm, Geometry, and Engineering Paradigm of China's Indigenous Closed Spherical Universe. Self-Preprint Archive. https://Zaitian001.github.io/-Self-Preprint-V1.1-/preprints/DW58553902.html

[3] Aristotle. (1831). Φ A [Physicorum libri VIII]. In I. Bekker (Ed.), Aristoteles Opera (Vol. 1, pp. 184–267). Academia Regia Borussica. (Original work published ca. 350 BCE)

[4] Ames, A. D., Coogan S., Egerstedt. et al. (2019). Control Barrier Functions: Theory and Applications. *2019 18th European Control Conference (ECC)*, 3420-3431. https://doi.org/10.23919/ECC.2019.8796030

[5] Yin Xi 尹喜. *Wenshi Zhenjing* 文始真经. (Original work published ca. 485 BCE)

[6] Shang Gao 商高. *Zhoubi Suanjing* 周髀算经. (Original work transmitted orally by Shang Gao and recorded in manuscript by Zhou Gong ca. 1046 BCE)

[7] Guoyu 国语. (Original work published ca. 455 BCE)

[8] Rovelli, C. (2018). The order of time. Riverhead Books.

[9] Zurek, W. H. (2003). "Environment-Assisted Invariance, Entanglement, and Probabilities in Quantum Physics."*Physical Review Letters* 90, 120404. https://doi.org/10.1103/PhysRevLett.90.120404

[10] Parhami, B. (2009). Computer Arithmetic: Algorithms and Hardware Designs. Oxford University Press.

[11] Murray, R. M., et al. (1994). A Mathematical Introduction to Robotic Manipulation. CRC Press.

[12] Goldstein, H., et al. (2002). Classical Mechanics. Pearson.

[13] Arnold, V. I. (1989). Mathematical Methods of Classical Mechanics. Springer.

[14] Baoliin (Zaitian) Wu 伍宝林. (2026). The Overlooked Coordinate System Exorcising the Demons and Ghosts of Quantum Measurement. Self-Preprint Archive. https://Zaitian001.github.io/-Self-Preprint-V1.1-/preprints/DW58553901.html

[15] Baoliin (Zaitian) Wu 伍宝林. (2026). Groma-Ju Dao: The Six Modes of the Zhoubi Suanjin "Ju" as a Geometric Operator System. Self-Preprint Archive. https://Zaitian001.github.io/-Self-Preprint-V1.1-/preprints/DW58553910.html

[16] Baoliin (Zaitian) Wu 伍宝林. (2026). Curvature-Driven Open Quantum Dynamics and the Pure Geometric Origin of Physics Constants. Self-Preprint Archive. https://Zaitian001.github.io/-Self-Preprint-V1.1-/preprints/DW58553914.html

[17] Baoliin (Zaitian) Wu. (2025). The Pure Geometric Origin of Mass and Light. Preprints. https://doi.org/10.20944/preprints202505.2249.v1

[18] Baoliin (Zaitian) Wu. (2025). The Return to the Einstein Spherical Universe Model. Preprints. https://doi.org/10.20944/preprints202501.2189.v1

[19] Baoliin (Zaitian) Wu. (2025). The Return to the Einstein Spherical Universe: The Dawning Moment of a New Cosmic Science. Preprints. https://doi.org/10.20944/preprints202504.0727.v2

2.17 Physical Proof of Existence (Appendix)

- **Banknote Serial:** `DW58553926`
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